The self-coupled dirac16complex in gravitational fields: symbols and their definitions

A dictionary of every symbol used for the nine-channel self-coupling, the Fierz matrix, the covariant nonlinear field equation, chirality in a gravitational field and the Kohn–Sham theory in the three epochs — sections 15–16 of the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one), the defining equation and its meaning, grouped by topic in the order of introduction. Longer defining equations follow each table as numbered displayed formulas. Every equation needed to define a symbol is written in this document.

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1. Conventions

- Units: $\hbar = c = 1$; a prime is d/dx_0.
- Coordinates $X = (x_0, x_1, \ldots, x_7)$ (`X = {x0, …, x7}`): x_0 time, x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space.
- Frame indices $A, B \in \{0, \ldots, 7\}$, raised and lowered with $\eta = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`); coordinate indices $\mu, \nu \in \{0, \ldots, 7\}$; $i \in \{1, 2, 3\}$ ordinary, $k \in \{4, \ldots, 7\}$ auxiliary space. Repeated indices are summed.
- Multi-indices K, J are strictly increasing lists of frame indices; $|K|$ is the rank (the length of K); $K \in r$ means $|K| = r$.
- Matrices: 1_n identity, M^{T} transpose, $M^{\dagger}$ conjugate transpose, tr trace, $[X, Y] = XY - YX$.
- Mathematica names with Greek letters are written with named characters: `\[CapitalTheta]up` is `Θup`, `\[Eta]K` is `ηK`, `\[Chi]17` is `χ17`, `\[Omega]17` is `ω17`.

2. Dirac matrices, chirality, and the field

symbol	Mathematica	definition	meaning
Γ_A	`Γ17[A]`	(2.1)	real 16×16 Dirac matrices of 7+1 spacetime
Γ^A	`Γ17up[A]`	$\eta^{AA}\Gamma_A$ (no sum)	
τ_i^O	`τO[i]`	real antisymmetric 8×8, $(\tau_i^O)^2 = -1_8$, anticommuting	the seven imaginary octonion units as matrices
ω	`ω17`	$\Gamma_0\Gamma_1\cdots\Gamma_7$	volume element; $\omega^2 = -1_{16}$
χ	`χ17`	$i\omega$	chirality; $\chi^2 = 1_{16}$, Hermitian
$P_{L,R}$	`PL17`, `PR17`	$\frac{1}{2}(1_{16} \mp \chi)$	chiral projections
E_L, E_R	`EL`, `ER`	(2.2)	chiral basis (16×8)
ϕ_L, ϕ_R	`phiLc`, `phiRc`	$E_{L,R}^\dagger\psi$	8-component chiral parts
ψ	`dirac16complex`; `psN`, `psR`	16 complex anticommuting components	the field; in the checks random Gaussian-integer spinors
$\bar\psi$	`pbS`, `pbR`	$\psi^\dagger\Gamma_0$ (independent symbols in the checks)	Dirac adjoint
α_j, β	`αD[j]`, `βD`	$\Gamma^0\Gamma^j$, Γ_0	Hamiltonian Dirac matrices
m	`mD`; `mks`		mass of dirac16complex ($m = 1$ in section 16)

$$\Gamma_0 = \begin{pmatrix} 0 & 1_8 \\ 1_8 & 0 \end{pmatrix}, \qquad \Gamma_i = \begin{pmatrix} \tau_i^O & 0 \\ 0 & -\tau_i^O \end{pmatrix}, \qquad \Gamma_A\Gamma_B + \Gamma_B\Gamma_A = 2\eta_{AB}1_{16}, \tag{2.1}$$

$$E_L = \frac{1}{\sqrt{2}}\begin{pmatrix} 1_8 \\ -i\,1_8 \end{pmatrix}, \qquad E_R = \frac{1}{\sqrt{2}}\begin{pmatrix} -i\,1_8 \\ 1_8 \end{pmatrix}, \qquad \chi E_L = -E_L, \quad \chi E_R = E_R. \tag{2.2}$$

3. Gravity

symbol	Mathematica	definition	meaning
${e^A}_\mu$, ${e_A}^\mu$	`fd["e"]`, `fd["ei"]`	$g_{\mu\nu} = \eta_{AB}{e^A}_\mu{e^B}_\nu$, ${e_A}^\mu{e^A}_\nu = \delta^\mu_\nu$	frame and inverse frame
$\Gamma^\mu(X)$	`fdCos["GamUp"]`	${e_A}^\mu\Gamma^A$	generalized frame matrices
$\{^\lambda_{\mu\nu}\}$	`cv["chr"]`	(3.1)	Christoffel symbols (never written Γ)
${\omega_\mu}^A{}_B$	`fd["om"]`	(3.2)	canonical spin connection
Ω_μ	`fdCos["Om"]`	(3.2)	spinor connection
D_μ	—	$\partial_\mu + \Omega_\mu$	covariant derivative of spinors
a, b	`a3[x0]`, `b4[x0]`; `aE`, `bE`	(3.3)	scale factors of ordinary and auxiliary space
H_a, H_b	`Ha`, `Hb`; `hE`, `kE`	a'/a, b'/b	Hubble rates
H_a', H_b'	`dHa`, `dHb`	dH_a/dx_0, dH_b/dx_0	
${E^{(k)\mu}}_\nu$	`ECos[[k+1]]`	(3.4)	Lovelock tensors ($E^{(1)}$ = Einstein tensor)
α_k	`alf[k]`	$\alpha_1 = 1$	Lovelock couplings (α_2, α_3)
$\bar\Lambda$	`lamb`		cosmological constant
κ	`kap`	$8\pi G_8$	gravitational coupling
p, q	`pKs`, `qKs`	$p = (1+2\sqrt{2})/7$, $q = (2-3\sqrt{2})/14$	Einstein–Kasner exponents: $a = x_0^p$, $b = x_0^q$

$$\{^\lambda_{\mu\nu}\} = \tfrac{1}{2}g^{\lambda\rho}\big(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}\big), \tag{3.1}$$

$$\omega_\mu{}^A{}_B = e^A{}_\nu\big(\partial_\mu e_B{}^\nu + \{{}^\nu_{\mu\lambda}\} e_B{}^\lambda\big), \qquad \Omega_\mu = \tfrac{1}{4}\eta_{AC}\,\omega_\mu{}^C{}_B\,\Gamma^A\Gamma^B, \tag{3.2}$$

$$ds^2 = dx_0^2 - a^2\big(dx_1^2 + dx_2^2 + dx_3^2\big) - b^2\big(dx_4^2 + \cdots + dx_7^2\big), \tag{3.3}$$

$$E^{(k)\mu}{}_\nu = -\frac{1}{2^{k+1}}\,\delta^{\mu\alpha_1\beta_1\cdots\alpha_k\beta_k}_{\nu\gamma_1\delta_1\cdots\gamma_k\delta_k} R^{\gamma_1\delta_1}{}_{\alpha_1\beta_1}\cdots R^{\gamma_k\delta_k}{}_{\alpha_k\beta_k}, \quad \delta^{\mu_1\cdots\mu_n}_{\nu_1\cdots\nu_n} = \det\big[\delta^{\mu_i}_{\nu_j}\big]. \tag{3.4}$$

4. The nine channels and the Fierz matrix

symbol	Mathematica	definition	meaning
K	`mIdx`	the 256 increasing subsets of $\{0,\dots,7\}$	multi-indices
$r = \lvert K\rvert$	`rk[K]`	length of K	rank, $0 \le r \le 8$
Γ^K	—	$\Gamma^{A_1}\cdots\Gamma^{A_r}$, $\Gamma^{()} = 1_{16}$	product of Dirac matrices
Θ^K	`θup[K]`	(4.1)	Hermitian-phased products; $\bar\psi\Theta^K\psi$ real
η_K	`ηK[K]`	$\eta_{A_1A_1}\cdots\eta_{A_rA_r}$	sign of the multi-index
Θ_K	`θdn[K]`	$\eta_K\Theta^K$	index lowered; $\Theta_K\Theta^K = 1_{16}$
I_r	`Ichan[r, pb, ps]`	(4.2)	Lorentz invariant of rank r
g_r	`gg[r]`, `gSymb`		the nine couplings, $r = 0,\dots,8$
g_S, g_V, g_A, g_P	`gS`, `gV`, `gA`, `gP`	$g_0 = g_S$, $g_1 = g_V$, $g_7 = -g_A$, $g_8 = -g_P$	the four couplings of section 8 of 7+1.nb
$\mathcal{L}_{\text{self}}$	`LselfS`	$-\frac{1}{2}\sum_r g_r I_r$	self-coupling Lagrangian
S	`sLL`	$\exp\big(\frac{1}{8}w_{AB}[\Gamma^A,\Gamma^B]\big)$, $w_{AB} = -w_{BA}$ real	local Lorentz transformation of spinors
F_{rs}	`fierzF`	(4.3)	9×9 Fierz matrix, $F^2 = 1$
R	`RrF`	a 16×16 matrix	density matrix (random in the checks)
D_s	`dS9[R]`	$\sum_{J\in s}\operatorname{tr}(\Theta_J R)\operatorname{tr}(\Theta^J R)$	direct (Hartree) invariants
G_s	`Geff`	$G = (1 - F^{\mathsf T})g$	Hartree–Fock (Fierz-transformed) couplings
G_S, G_V	`gSks`, `gVks`	$G_S = G_0$, $G_V = G_1$	couplings of uniform matter (−100000, 10000)
ρ_S, ρ	`rhoSs`, `rhos`	$\bar\psi\psi$, $\psi^\dagger\psi$ (expectation values)	scalar and number densities

$$\Theta^K = i^{\sigma_r}\,\Gamma^{A_1}\cdots\Gamma^{A_r}, \qquad \sigma_r = \tfrac{r(r-1)}{2} \bmod 2, \tag{4.1}$$

$$I_r = \sum_{|K|=r} (\bar\psi\Theta_K\psi)(\bar\psi\Theta^K\psi), \tag{4.2}$$

$$F_{rs} = \frac{1}{16}\sum_l (-1)^{rs-l}\binom{s}{l}\binom{8-s}{r-l} = \frac{1}{256}\sum_{K\in r}\operatorname{tr}\big(\Theta_K\Theta_J\Theta^K\Theta^J\big) \quad (|J| = s). \tag{4.3}$$

The Fierz identity for commuting spinors a, b, c, d:

$$\sum_{K\in r}(a\Theta_K b)(c\Theta^K d) = \sum_s F_{rs}\sum_{J\in s}(a\Theta_J d)(c\Theta^J b).$$

5. Field equation and Kohn–Sham quantities

symbol	Mathematica	definition	meaning
$\Sigma(X)$	—	(5.1)	Kohn–Sham self-energy (direct minus exchange)
φ_n	—	$(i\Gamma^\mu D_\mu - m - \Sigma)\varphi_n = 0$	Kohn–Sham orbitals
$\mathcal{R}$	—	$\sum_{n\ \mathrm{occ}} \varphi_n \bar{\varphi}_n$	Kohn–Sham density matrix
$k = (\vec{p}, \vec{q})$	—	comoving momentum	$\vec{p}$ ordinary, $\vec{q}$ auxiliary
k_F	`kF`	radius of the comoving Fermi ball	Fermi momentum
$k_{F,\mathrm{sat}}$	`kFsat`	$p = 0$ at $a = b = 1$	saturation Fermi momentum
θ	—	$\lvert k_a\rvert = k\cos\theta$, $\lvert k_b\rvert = k\sin\theta$	angle between ordinary and auxiliary momentum
ω (energy)	`om`	$\sqrt{k^2\cos^2\theta/a^2 + k^2\sin^2\theta/b^2 + M^2}$	single-particle energy (not the volume element)
M, m^*	`Mx`, `ksGap`	$M = m + G_S\rho_S(M)$	effective mass (gap equation)
V_0	—	$G_V\rho$	vector potential
$\langle f\rangle$	`ksMoments`	(5.2)	average over the comoving Fermi ball
ε, p_a, p_b	`ksStress`	(5.3)	energy density, pressures along ordinary / auxiliary space
Π	`pint`	(5.3)	interaction pressure
ε_F	—	$G_V\rho + \sqrt{k_F^2 + M^2}$	Fermi energy (flat spacetime)
E/A	—	ε/ρ	energy per particle
$T^\mu{}_\nu$	—	$\mathrm{diag}(\varepsilon, -p_a, -p_a, -p_a, -p_b, \ldots, -p_b)$	energy–momentum tensor of the Kohn–Sham matter
—	`evolveEL[α2, α3, κ, Ha0]`	returns $\{H_b(0), \{a, b, H_a, H_b\}(20),$ violation$\}$	self-consistent Einstein–Lovelock + Kohn–Sham evolution

$$\Sigma = \sum_r g_r \sum_{|K|=r} \big[\mathrm{tr}(\Theta_K\mathcal{R})\,\Theta^K - \Theta^K\mathcal{R}\,\Theta_K\big], \tag{5.1}$$

$$\langle f\rangle = \frac{1}{2\pi^4 a^3 b^4}\int_0^{k_F} dk\,k^6 \int_0^{\pi/2} d\theta\,\cos^2\theta\sin^3\theta\,f, \qquad \rho = \langle 1\rangle, \quad \rho_S = \langle M/\omega\rangle, \tag{5.2}$$

$$\varepsilon = \langle\omega\rangle - \frac{(M-m)^2}{2G_S} + \frac{G_V}{2}\rho^2, \quad \Pi = \frac{(M-m)^2}{2G_S} + \frac{G_V}{2}\rho^2, \tag{5.3}$$

$$p_a = \Big\langle \frac{k^2\cos^2\theta}{3a^2\omega}\Big\rangle + \Pi, \quad p_b = \Big\langle \frac{k^2\sin^2\theta}{4b^2\omega}\Big\rangle + \Pi.$$

6. Symbols that look alike

symbol	not to be confused with
$\omega = \Gamma_0\cdots\Gamma_7$ (volume element)	ω (single-particle energy), $\omega_\mu{}^A{}_B$ (spin connection)
χ (chirality)	χ_k (rescaled mode, $\psi = (a^3b^4)^{-1/2}\chi_k$)
Θ^K (upper multi-index)	$\Theta_K = \eta_K\Theta^K$ (lower multi-index)
g_r (bare couplings)	G_s (Fierz-transformed couplings), $g_{\mu\nu}$ (metric)
F_{rs} (Fierz matrix)	$\mathcal{F}_{\mu\nu}$ (spin curvature, not used here)
R (density matrix)	$\mathcal{R}$ (Kohn–Sham density matrix), $R^{\gamma\delta}{}_{\alpha\beta}$ (Riemann tensor)
p (Kasner exponent; pressure in flat spacetime)	p_a, p_b (pressures), $\vec{p}$ (ordinary momentum)

symbol	**not to be confused with**
q (Kasner exponent)	$\vec{q}$ (auxiliary momentum)
α_k (Lovelock couplings)	α_j (Dirac matrices $\Gamma^0\Gamma^j$)
$\kappa = 8\pi G_8$	—